

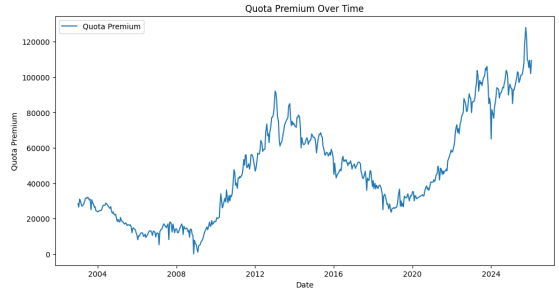
Predicting Certificate of Entitlement Price in Singapore

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What is COE?

- The Certificate of Entitlement gives you the right to own a car in Singapore
- Constitutes the bulk of a car's price
- COEs are distributed through an open-bid auction system, which happens twice a month



- Bidders put in their maximum willingness to pay (WTP)
- COEs are allocated from the highest bidder downwards
- In the end, everyone pays the same Quota Premium (QP), which is the WTP of the highest unsuccessful bidder plus \$1
- In practice, car dealers do the bidding

Research Questions

- 1 Can COE prices be predicted better than standard benchmark models (e.g., random walk)?
- 2 Do machine-learning gains in forecast accuracy (if any) translate into higher business profits?

Data and Features

- Sample: Cat A COE prices from Jan 2003 to Jan 2026 (548 observations)
- Target modeled in growth rates, then reconstructed to level QP
- Features include supply, demand, policy, market, and macro proxies (55 features)
- Each feature and target series is lagged 12 times

Modeling Workflow

- Train/test split: last 96 observations reserved for testing.
- 3-fold expanding-window time-series cross-validation for hyperparameter tuning. Grid search used.
- One-step-ahead walk-forward forecasting on the test set.
- Models: Random Walk, Auto-ARIMA, Elastic Net, SVR, XGB, XGB_custom, Chronos-2.
- Other than XGB_custom, hyperparameters are tuned with MAE as loss metric. XGB_custom tuned on custom profit metric
- Implemented with Darts

- Firm, at time $t - 1$, price cars based on prediction, plus margin: $\text{Price}_t = \text{Pred}_t + \text{margin}$
- Firm guarantees buyer that COE will be obtained within 2 biddings
- At time t , bid whatever model predicts. If win, profit = $\text{Price}_t - \text{QP}_t$
- If lose at time t : secure at $t + 1$ by bidding very high, profit = $\text{Price}_t - \text{QP}_{t+1}$
- Demand (probability of buying) is modeled with a logistic function anchored to the latest QP.

$$\text{Demand}_t = \frac{1}{1 + e^{k(\text{Price}_t - \text{QP}_{t-1})}}$$

Results

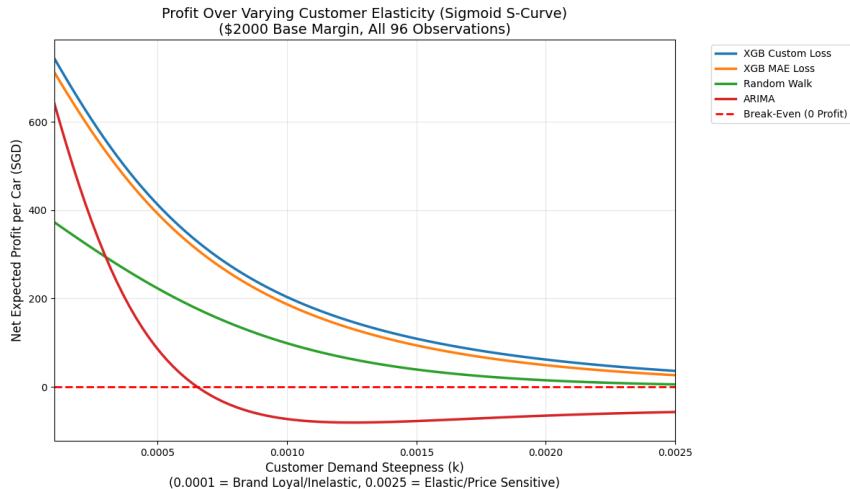
- XGB performs best
- Diebold–Mariano test: XGB beats Random Walk (p-value: 0.02) but not AutoARIMA (reported p-value: 0.50).

Model	RMSE	MAE
Random Walk	5210	3823
AutoARIMA (MA2)	5415	3761
Elastic Net	5263	3680
XGB	5046	3622
SVR	5230	3755
Chronos-2	5421	3968
XGB_Custom	5048	3647

Test-set forecasting performance

Profit metric

- The best models XGB and XGB_custom are further evaluated on the profit metric.



Profit metric

Metric	XGB vs RW	XGB_custom vs RW
Profit difference	169.00 (76% increase)	189.56 (85% increase)
p-value	0.0062	0.0045

Profit-differential DM-test against Random Walk

Conclusions

- XGB can exploit covariate information to achieve modest improvement in predictions
- More importantly, even modest improvements can translate to large increases in expected profit
- Future work: change to a classification task/benchmarking against firm's actual performance.

Thank you!

Feature	Correlation
quota_premium_cat_b_growth_rate	0.568
overdemand_growth_rate	0.462
overdemand_cat_b_growth_rate	0.310
quota_premium_growth_rate_lag_1	-0.262
overdemand	0.259
cat_a_divide_cat_b_lag_1	-0.216
quota_per_bid	0.006
quota_per_bid_growth_rate	-0.047

Selected feature correlation with QP growth rate

Definitions: $\text{overdemand} = (\text{bids received})/(\text{quota})$; quota_per_bid = approximate quota announced before bidding.

Variable importance

Base feature	SHAP value	Abs. SHAP
QP growth rate	0.57	1.21
cat_a_divide_cat_b	0.57	0.60
overdemand_cat_b_growth_rate	0.13	0.59
overdemand_cat_a_growth_rate	-0.18	0.37
deregis_cat_a_growth_rate	-0.23	0.28

SHAP values for the last test-set prediction

Figure A1: Actual vs prediction

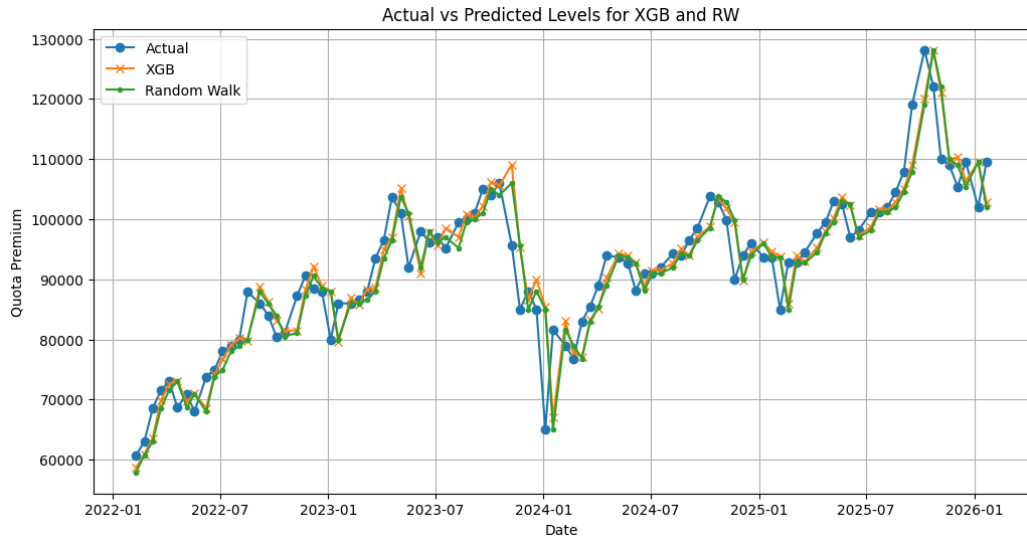


Figure A2: Robustness to margin values

